

B. Wire Rod Block Mill – 6 Stand**WRM Design Basis**

Proposed equipment layout is indicated in drawing. Product which can be rolled through this line will be wire rods from dia **5.5mm to dia 8mm** in coil form.

Process

- Rolling stock from the last stand of the continuous mills will be directed to a **6 stand** “Twist free” block mill for producing finished Wire Rod
- At the entry, a horizontal pinch roll will be located which will assist the bar in travelling to block mill.
- Downstream, a disc & drum shear is located to crop the nose & tail of bar before entry to block. This shear will also cut the bar into fragments in the event of a cobble occurring downstream in the mill.
- After the shear, a side looper will be located to create a loop before the block mill. This loop is required to maintain the stock size at the block mill. Scanner located on this looper will be used to control the bar mill rolling speed to control the loop. Block mill speed will be fixed and speed of mill upstream will be varied.
- The twist free finishing block is of an advanced design, engineered to meet the strenuous demands of today’s high speed rod mills. The configuration comprises an electric motor driving through a synchronizing gearbox to give twin output extended to bevel gearboxes to from the two main mill drive-shafts of the block. From these drive shafts the bevel gears take the drive to the stands at 45o on each side of the vertical centerline so giving 90o stand to stand relationships. The driveshaft’s and bevel boxes etc, is all below pass line thus giving a low overall mill height for maximum operator access, visibility and operation. The mill is designed and built on a modular design and carry all rolls and rolls holders of identical dimensions and all entry and delivery guide holder are identical on all stands.

The roll / roll holders are mounted onto the roll shafts by means of hand operated hydraulic mounting tools and are dismounted using similar units. This allows rapid roll changing giving positive benefits in production time. The majority of stands having two pass grooves on the rolls and as the roll / roll holders are reversible, a further saving in carbide utilization is achieved.

- The stiffness of the stand, enabling it to roll closed tolerance, is gained from design features limiting the rolling loads stress path to a minimum and by pre-stressing the unit by means of a hydraulic cylinder. The design and cylinder facility also allows infinite roll center setting and adjustment within the maximum to minimum roll diameter range of 216-205 nominal. Axial adjustment is also provided on one roll shaft to allow accurate alignment of roll grooves.



- Finished stock from block mill will pass through a water cooling unit to reach the horizontal coil laying head. Flow of water through this unit is controlled for rapid cooling to drop the laying temperature of rolled plain bars to desired level.
- Plain wire rod laid in a spiral form by horizontal coil laying head on the air cooling conveyor will reach the coil reform station and gets formed in to a coil on the mandrel of a retractable type coil mandrel located below reform tub.
- Motor driven roller table will transfer the loosely formed coil on to the coil mandrel to a on line coil compactor before shifting to a 4 arm coil capstan.
Compacted coils from coil capstan need to be shifted to storage bay either by an overhead crane or by fork lifts.

SAMPLE ROLLING PARAMETERS

No	Bar size mm	Max. Bar speed m/s	Production Tons/hr.	No. of Stands
1	Dia. 5.5	60	30	6
2	Dia. 6	50	30	6
3	Dia. 8	30	30	2

GRADE OF STEEL ROLLED

Low Carbon steels.



CONTROLLED COOLING

Finished wire rod from Twist free block mill and laid on the roller conveyor in spiral layers by Coil laying head will be cooled in a controlled manner on the roller conveyor located between coil laying head and coil reforming station.

Cooling of wire rod will be in following ways depending on requirement of grade of steel;

- **Natural cooling:** Wire rod will be cooled with natural air between coil laying head and coil reforming station
- **Fast cooling :** Air blowers located below the conveyors will force the air to cool the wire rod. Cooling rate will be controlled by controlling the air quantity / using limited number of blowers
- **Retarded cooling :** Retractable Insulated hoods mounted on the roller conveyor frames will cover the conveyor upper side to retard the natural cooling rate.

No	Type of cooling	Grades of steel
1	Fast Cooling qualities	Typical S.W.P, Tyre bead, plain carbon of 0.3 to / 0.85 % C content
2	Slow Cooling qualities Typical	E.W.R, SAE 8620, EN45, 65Si7, 20Mn Cr5,/ S.A.E 52100, 16 MnCr5, 0.23%C



2. Equipment Technical Details**2.1 Pinch Roll**

Pinch roll unit is installed to assist the rolled bar travel from intermediate stand to the block mill.

The equipment consists of a set of rolls made of wear resistant casting, mounted on shafts supported on antifriction bearings. Top and bottom roll bearing housings are located on fabricated links pivoted on pivot frame. Top roll link is connected to a pneumatic cylinder which will control the pinching pressure of rolls on the bar.

Drive to the rolls from Motor is transmitted through a pinion gear box with hardened and ground gears. Gears and bearings of the gear box are splash oil lubricated. Pinch roll unit, gear box and motors are mounted on a common base frame.

Type	Cantilever type
Roll dia	250 mm Nominal
Roll material	Wear resistant cast iron
Drive Method	By electric motor through pinion gear box & propeller shafts
Pinching action	Pivoted top roll actuated by a pneumatic cylinder
Motor	30kW, 1500 RPM DC
Motor Control	DC variable speed drive
Speed	25 m/sec max. & 10 m/sec min.
Bar Size	Dia. 8mm to Dia.25mm
Gear Lubrication	Splashed oil lubrication

2.2 Crop & Cobble Shear

Type	Disc & Drum Shear
Max. bar Dia	13mm
Max. bar Speed	22 m/sec
Shear Blade Centres	300mm
Motor capacity	11 kW, AC / 15 KW AC
Blade Hardness	50 (+/- 2) HRC
Blade Material	Ni-Cr-Mo Steel

2.3 Side Looper

Type	Horizontal
Swing Arm	Pneumatic Operated
Max. Bar Speed	23 m/sec



2.4 6 Stand Twist Free Finishing Block Mill

The finishing Mill will comprise 170 / 155 mm cantilever type stands, arranged in a 45° opposed configuration, being alternately right and left hand.

Specifications:

No. of Stands	6
Angle of roll axes	45° V Type No-Twist
Max design speed	60 m/sec
Product size range	5.5mm to 8mm
Maximum Input size	11mm
Grades to be Rolled	Low Carbon Steel
TC Ring Size	OD 170mm max & 155mm min.
TC Ring Weight	11.7 kg approx.
Width & Number of pass per roll	70mm, 2 passes per ring
Roll Neck Dia	140mm
Roll balancing	Hydraulic cylinder accumulator @ 200kg/cm ²
Drive	DC Motor 750 kW 0-600-1200rpm
Lubrication System Capacity	700 lpm
Water flow Rate for Ring Cooling	200 m ³ /hr at pressure of 4-8kg/cm ²
Air Flow Rate	200 m ³ /hr
Average Reduction per pass	19%
Roll Stand Front Plate & Guides	Stainless Steel



Construction:**Bevel Gear Box**

- Drive to rolls will be transmitted through a bevel gearbox
- Bevel gearbox equipped with forged steel precision cut hardened and profile ground Spiral bevel gears
- The input and output shaft will run in cylindrical roller bearings and Angular contact ball bearings
- Gears and bearings will be arranged for oil lubrication from a Centralized oil system

Gears	Single helical dynamically balanced
Gears rating	400kW
Quality	En353 Case Hardened & Ground 60 +/- 2 HRC
Lubrication	Forced

Roll Stand

- Fabricated steel box housing of robust construction
- Forged steel single helical top and bottom roll shaft carried in cast steel gear links within the housing
- The roll shaft will run in white metal bearings at the roll end and roller bearing at the other, a double angular contact bearing being employed to accommodate axial loads
- Forged steel single helical drive shaft carried in the links will transmit the drive to the roll shafts
- The drive shafts will run in roller bearings with “four point” ball bearing taking the gear thrust loads
- The gear and bearing will be oil lubricated from a centralized oil system
- The roll neck labyrinth sealing will be supplemented by air / oil lubrication
- The roll gap will be manually adjusted by means of synchronized Twin screw and wedge arrangement acting on the gear links
- The links will be maintained in contact with the wedge gear by a hydraulic pre-stressing cylinder
- Axial roll adjustment gear for roll groove alignment will be In-corporated on the top roll shaft
- The adjustment gear will be manually operated

Size	216mm
Gears	Single helical dynamically balanced
Gears rating	400kW
Quality	En353 Case Hardened & Ground 60 +/- 2 HRC
Front Seal Plate & Covers	Stainless Steel
Gap Adjustment	Synchronised twin screw & wedge
Roll Balancing	Hydraulic cylinder at 200 kg/cm ²
TC Ring mounting	Held via taper sleeve pressed by hydraulic mounting unit
Axial Adjustment	+/- 2.0mm online without packing / removal of ring
Roll cooling	Stainless steel nozzles with flat / round spray



Synchronizing Gearbox

- The drive will be transmitted to the individual stand drive bevel gearbox through a common synchronizing gearbox
- Fabricated steel gearbox equipped with forged steel through hardened single helical gearing and twin output shafts
- Gears and bearings will be arranged for oil lubrication from a centralized oil system
- Steel cordon shafts equipped at each end with a flexible gear type
- Coupling will be provided from the synchronizing gearbox to the bevel gearbox, from bevel gearbox to bevel drive gearbox
- For each bevel drive gearbox, a double engagement flexible gear Type coupling will transmit the drive to the individual stands

Gears	Single helical dynamically balanced
Gears rating	4000kW
Quality	Case Hardened & Ground 60 +/- 2 HRC
Lubrication	Forced

Oil Lubrication System

Centralized oil lubrication system to serve the No-twist mill. System will comprise the following :

- One tank complete
- Two positive displacement pumps (1 running, 1 standby. Each pump will be complete with a bed-plate, motor coupling and coupling guard
- Suction strainers
- Direct acting relief valve for pump protection
- In line filters having manual changeover facility, giving 20 micron filtration
- Low level alarm in tank
- All necessary pipe-work, valves, switches, indicators and instrumentation at the pumping station only

Description	Block Lubrication
Pump Type	Screw/Gear Pump
No. of Pumps	1 + 1 standby
Flow & Pressure	600 lpm at 6 bar
Filtration Level	20 micron
Tank Capacity	6000 litres
Heat Exchanger	Plate Type

Centrifuge

One centrifuge will be provided for the purification of lubrication oils by removal of solids and water. Capacity is 400 lph and filtration level is 5 micron.

Emergency Pressure Vessel

An emergency pressure vessel of 1000 liters capacity will be provided to give approximately 2 minutes lubrication to the finishing mill in the event of a power failure.



ARRANGEMENT OF OIL LUBRICATION SYSTEM (TYPICAL ONLY)**Pre-Stressing Hydraulic System for Roll Balancing**

One (1) hydraulic system to serve the pre-stressing cylinder on the finishing block.

- Hydraulic medium : mineral oil
- Working pressure : 200 bar

System will comprise the following:

- One tank complete having a capacity of 150 liters
- Two positive displacement pumps (1 running, 1 standby) each having a delivery of 4.5 liters per minute at 200 bar. Each pump will be complete with a bell housing & motor coupling
- One relief valve for system overload.

Protection Cover Hydraulic System

One (1) hydraulic system to serve the cylinders for raising / lowering the hoods for the block mill stands. Hoods can be individual for each stand or common.

- Hydraulic medium : mineral oil
- Working pressure : 250 bar

System will comprise the following:

- One tank complete having a capacity of 150 liters
- Two positive displacement pumps (1 running, 1 standby) each having a delivery of 4.5 liters per minute at 10 bar. Each pump will be complete with a bell housing & motor coupling
- One relief valve for system overload.

Guides (Stainless Steel)

The followings guides will be provided for the finishing mill

- Roller entry guides with air oil lubrication for bearings.
- Exit stripper guides.

All Entry Roller Guides, Static Exit Guides, Guide Holders & Dummy Tubes will be made of Stainless Steel



Cobble Block

A cobble block will be provided immediately after the last stand finishing block to provide a 'break-out' point should a cobble occur downstream finishing line. Stainless steel cobble block with hinged, loose fitting top section will be located within the finishing block cobble guard.

Roll Holder

The roll holder will be mounted onto the roll shaft by means of tapered sleeves driven on by a hydraulic unit and removed by a similar dismounting unit

Roll Mounting & dismounting equipment

- The equipment will be designed for the mounting and dismounting of Roll and roll assemblies from the roll shaft of the stand.
- Roll assemblies will be secured to the roll shaft by tapered locking sleeves Hydraulically driven into the bores of the rolls and roll assemblies
- A specially adapted hydraulic unit will be provided for this purpose together with a similar unit for removal of the tapered sleeves

Portable & Bench Optic System

One portable & one bench optic will be provided for setting of guides on block mill & in guide shop

Roll Unit Changing

Since Roll unit & bevel units are independently enclosed and connected by gear coupling, separate tackle is not required. Hooks are provided on stands for lifting by crane. Roll unit can be opened separately in dust free room without risk of contamination on field

Air Oil Lubrication System

Air Oil lubrication system provides air oil mist for guides and also seals of roll stands to prevent water contamination. This improves life of seals also. Timing of this system is PLC Controlled

Temperature Indication of Bearings

Temperature of bearings will be displayed in Pulpit by connecting bearing RTDs to analog inputs of the main PLC. Alarm & Tripping will be done by system in case of overheating.



2.5 High speed Pinch Roll (1 no.)

Located before the coil laying head, the unit assists rod entry into the coil laying head.

Design

Diameter of rolls :	300mm
Rod Size :	5.5mm to 8mm Dia
Motor :	60 KW, 1500 RPM, AC

Technical Specifications

Fabricated steel box housing of robust construction. Forged steel single helical top and bottom roll shafts carried in fabricated steel links within the box housing. Forged steel single helical drive shaft carried in the links will transmit the drive to the roll shaft. Roll shafts and drive shafts carried in anti-friction bearings. Each link will be connected to a crank sleeve mounted within the housing. A pneumatic cylinder rotates both crank sleeves to pivot the link and bring the pinch roll together to provide the pinch.

Drive

Pinch roll unit driven by AC electric motor through a flexible gear type coupling. One drive shaft is extended through the housing to form the input shaft. Drive shaft will be meshed together to transmit the drive one to the other.

Rolls

Wear resistant graded cast iron rolls mounted on to the tapered roll-shafts

Pinch Mechanism

A pneumatic cylinder is directly connected to the crank sleeve and by an extension rod to the bottom crank sleeve. Adjustment to remove backlash from the system is provided within the extension rod. The roll travel to provide the pinch is governed by fixed length stroke of the pneumatic cylinder.

Adjustment of the position of the cylinder and therefore open dimension of the pinch rolls is used along with the fixed cylinder stroke to adjust the final roll gap when pinched. Cylinder position adjustment is by a ratchet mechanism external to the pinch roll housing.

Lubrication

The bearings and gears of the pinch roll unit is oil lubricated from lubrication system of block mill.



2.6 Coil Laying Head :**Function**

The coil laying head receives rod from the water boxes / pinch roll unit and lays it in an overlapping spiral configuration onto the coil conveyor

Design

Diameter	:	Nominally 1050
Balance	:	ISO 1940 – 1973 Grade 25
Angle of Inclination	:	15°
Rod Sizes	:	Dia. 5.5mm to Dia. 8mm
Motor	:	75 KW, 1500 RPM

The laying head consist of a rotating shaft to one end of which is attached a rotor which support the formed laying pipe. The rotating shaft is hollow and the rod is fed through the center of this shaft up to the laying pipe by a water cooled tubes.

Drive

The laying head is driven by a AC electric motor connected to the laying head input shaft by a flexible coupling. Drive is transmitted from the input shaft to the laying shaft by helical gearing.

Laying Pipe

The laying pipe is in single piece mounted on to coil cone with fabricated clamps.

Bed Plate

Bed-plate of fabricated steel to support the coil- laying head. The laying head motor is mounted directly onto the back of the laying head gearbox base frame.

Construction

The laying head casing and laying pipe guard is of fabricated steel. Laying shaft and rotor of alloy steel. Helical gearing of alloy steel. The input shaft and laying shaft are supported in anti-friction bearings.

Lubrication

The bearing and gear of the laying head are oil lubricated from lubrication system of block mill.



2.7 Laying Pipe Bending Jig

Function

For the manufacture of replacement laying pipes.

Design

The new pipe is located in the bottom of jig, heated locally, and bent around the pre-formed shape of the jig. The pipe is clamped to the as it is formed.

Construction

Fabricated steel bending jig with machined support positions. Steel clamps to hold the pipe in position during bending.



CONTROLLED COOLING EQUIPMENT**2.8 Coil Cooling Conveyor:**

Type	Roller Conveyor
Roller Conveyor length	30 meters
Roller Diameter	100mm (solid steel rolls for first segment of 4 meters) Hollow rollers there after
Roller length	1400mm
Roller Pitch	approx. 105mm at laying head and 200mm there after
Bearings	Pillow blocks
Number of drives	Five
Drive system	AC geared Motor , sprocket and chains
Drive motor	AC variable speed
Rating	11kW 625 rpm
Chain	1 inch pitch, single chain

Function

The coil-cooling conveyor will transport and advance the open laid coil of rod issuing from the coil laying head through the air-cooling section and upto the reform tub.

Design

- Overall conveyor length 65 meters
- Roller Conveyor
- Solid and pipe steel rollers carried in pillow blocks
- Fabricated steel conveyor frames supports bearings
- Rollers interconnected with chain and sprockets
- chain guard of fabricated steel
- Stub rollers at the step end of the conveyor
- Adjustable Section at the laying Head
- A section of conveyor at the coil laying head will be hinged and may be elevated or lowered to provide access for laying pipes removal, and to obtain optimum rod ring formation on the conveyor

Roller Section at Reform Tub

- Generally as the main roller conveyor
- Stub rollers at the reform tub
- Drive taken by transmission chain from the chain conveyor
- Adjustable longitudinally to provide optimum rod ring presentation into the reform tub
- Adjustment is manual

Adjustable Side plates

Manually adjustable side plates will be provided at the step between the roller conveyor and the chain conveyor and at the reform tub.



2.9 COIL FORMING AND HANDLING EQUIPMENT**2.9.1 REFORM STATION:**

One (1) fabricated steel reform tub mounted at the end of cooling conveyor on fabricated steel support structure

The coil plate will be driven by a hydraulic cylinder

Also incorporated will be hydraulic cylinder operated reform gate which will control the outside diameter of the coil

A fabricated steel pilot cone will be provided in the reform tub which will aid coiling of the rod whilst awaiting mandrel availability

The cone will rest loosely on the end of the mandrel when it is in a vertical position within the reform tub

The reform tub will receive coil spirals from the cooling conveyor and direct them on to the disappearing mandrel.

2.9.2 RING DISTRIBUTION SYSTEM (Optional)

Ring distribution system ensures distribution of rings to enable smaller coil height for given weight. Ring distribution system will be of either slewing bearing type operated by geared motor or variable protrusion type

2.9.3 COIL IRIS PLATE & CONE

4 nos of coil iris plate will be on either side of the cone. The plates will be driven by a hydraulic/ pneumatic cylinder. A fabricated steel pilot cone will be provided in the reform tub which will aid coiling of the wire rod whilst awaiting mandrel availability. The cone will rest loosely on the end of the mandrel when it is in a vertical position within the reform tub. Coil Plate opens and rests the cone on the mandrel

2.9.4 DISAPPEARING MANDREL WITH EASY DOWN

A disappearing mandrel surrounded by a cage will receive the coil on the roller conveyor. A vertically moving mandrel to support the Nose Cone for Coil Inside diameter control hydraulically operated. Along with the centring mandrel, a coil receiving mechanism called Easy Down mechanism supports the coil and slowly brings it down as the coil height builds up. This helps in maintaining a proper coil shape which is easy to compact, bind and transport. (The mandrel disappears / comes down when this easy down mechanism comes to its lowest level on the roller conveyor.) The easy down mechanism works hydraulically and is located inside the mandrel / boom. Since it goes through the roller conveyor, coil cage has its arms extended in only the axis which is parallel to the roller conveyor.

2.9.5 ROLLER CONVEYOR

Roller conveyor is located below the reform station. It will convey vertical coil collected in mandrel and passes it online coil compactor. Length is approx 6.5m. Roller conveyor is operated by geared motor.



2.9.6 ONLINE UP COMPACTOR

Online Up-compactor will lift and compress coil upto 1400kg. This operation is done by hydraulic cylinders. During this period, coil will be strapped manually before moving the coil ahead.

2.9.7 DELIVERY ROLLER TABLE

Delivery Roller Table is located below and after compactor. Delivery Roller Tables transfers compacted tied coils to Upender through geared motor.

2.9.8 WEIGHING STATION (Optional)

One delivery table is suspended on load cells for online coil weighing & printing for the tags

2.9.9 UP ENDER KICKOFF

Travelling type Up-ender lifts the compacted tied coil and kicks it off below or places it on the arms of the capstan.

2.9.10 COIL CAPSTAN (Optional)

The coil capstan has 4 arms to hold the compacted coils for further processing.

